

LUYI TONG

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EDUCATION

University of Washington | Seattle, United States
M.S., Biostatistics | Sept 2024 – Mar 2026

McGill University | Montreal, Canada
B.A., Honours Probability and Statistics | Sept 2020 - June 2024
Minor: Economics

CORE SKILLS

- **Programming:** Python (Pandas, NumPy, Scikit-learn), R, SQL, Java
- **Data Engineering:** PySpark, Hive, ETL, Data Pipeline, Workflow Automation, Feature Engineering, Data Cleaning, Data Quality Monitoring
- **Analytics:** A/B Testing, Retention Analysis, Predictive Modeling, Machine Learning, Model Validation, Cohort Analysis, Customer Segmentation, Forecasting, Causal Inference
- **Visualization:** Power BI, Tableau, Dashboard Development, Data Visualization, KPI Monitoring

EXPERIENCE

Data Analyst Intern

Future Force — Seattle, United States

Jun. 2025 – Sept. 2025

- Analyzed structured and unstructured datasets to identify high-impact business opportunity signals, **increasing opportunity coverage by 20%** while improving signal quality across analytical workflows.
- Automated data pipelines and scoring frameworks using Python and SQL to transform, validate, and prioritize opportunities based on impact and feasibility.
- Conducted user behavior and engagement analyses for AI-enabled eldercare platform, leveraging feature engineering and then monitoring, to identify caregiver burden patterns, **increasing user engagement by 10%**.
- Built and validated predictive risk-scoring models to identify vulnerable user groups, improving early detection of high-risk cases by **12%**, for supporting proactive intervention strategies.
- Created Power BI dashboards, establishing KPI monitoring and delivering actionable insights.

Digital Marketing & Data Analytics Intern

Ningbo Chunqiu Guantu Culture and Media Co., Ltd — Zhejiang, China

- Built and maintained ETL pipelines integrating data from 20+ marketing channels using SQL, PySpark, and Hive.
- Automating daily data ingestion from multiple APIs, supporting dashboard reporting and performance monitoring.
- Developed attribution models to evaluate channel effectiveness and optimize budget allocation, **increasing campaign ROI by 25%**, through data-driven investment decisions.
- Performed customer segmentation and behavioral analyses to identify high-value user groups, **improving retention by 15%** through targeted engagement strategies.
- Developed forecasting models and KPI monitoring dashboards that reduced analytical turnaround time from **7 days to 72 hours**, enabling faster cross-functional decision-making.
- Visualized analytical findings on conversion, retention, and campaign effectiveness to support growth strategy and customer engagement initiatives.

PROJECT WORK

Conversion Optimization Analysis

Technologies: SQL, Python, Data Visualization, Segmentation, A/B Testing

- Extracted, cleaned, and transformed customer transaction and behavioral data using SQL and Python, resolving data quality issues and preparing datasets for downstream analysis.
- Developed funnel analysis frameworks to evaluate user conversion across browsing, add-to-cart, checkout, and payment stages, identifying major drop-off points and conversion bottlenecks.
- Performed RFM-based customer segmentation and behavioral analysis to identify high-potential new customers and churned high-value users, enabling targeted engagement strategies.
- Designed and evaluated A/B tests for promotional campaigns and customer reactivation initiatives, improving **overall conversion rate from 0.8% to 1.6%**.

A/B Testing / Experimentation Toolkit

Technologies: Statistical Testing, Experiment Design, R Package Development

- Developed an R package to simulate A/B testing experiments with binary outcomes and compare success rates between treatment and control groups.
- Implemented statistical testing functions to calculate conversion rate differences, p-values, and confidence intervals for experiment evaluation.
- Built visualization functions using ggplot2 and documented the package with examples, tests, and a pkgdown website to improve usability and reproducibility.